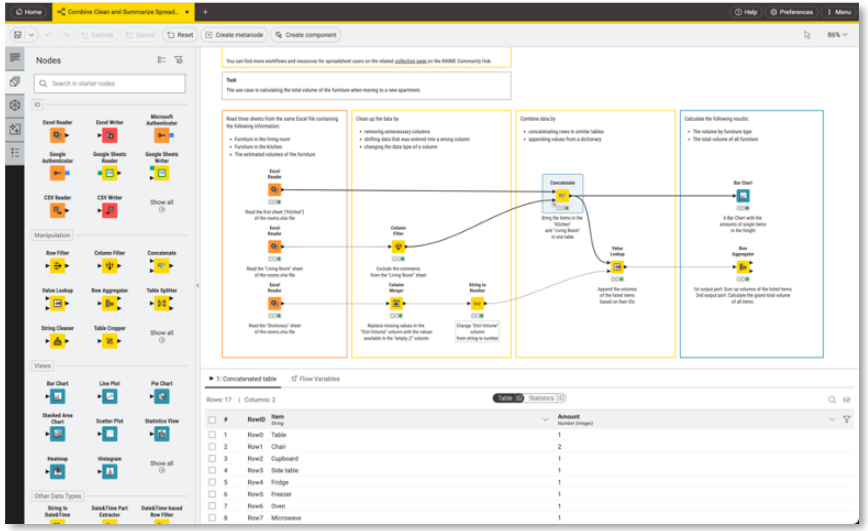
Introdução à Plataforma KNIME Analytics

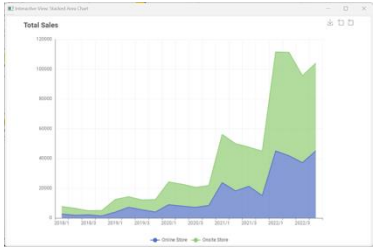
Fatec

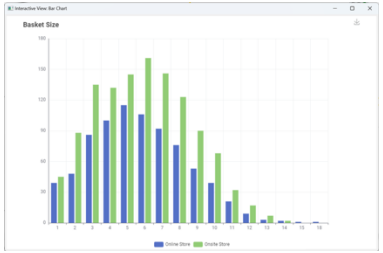
2026

O que é a Plataforma analítica KNIME

- Ferramenta para dar sentido aos dados
  - Data analysis, Data Science, Data Engineering
  - Transformation, visualization, reporting
- Open source
- Paradigma de programação Visual
  - Não é necessário codificar







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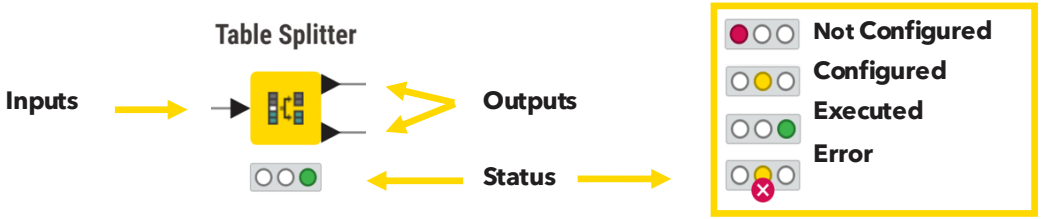
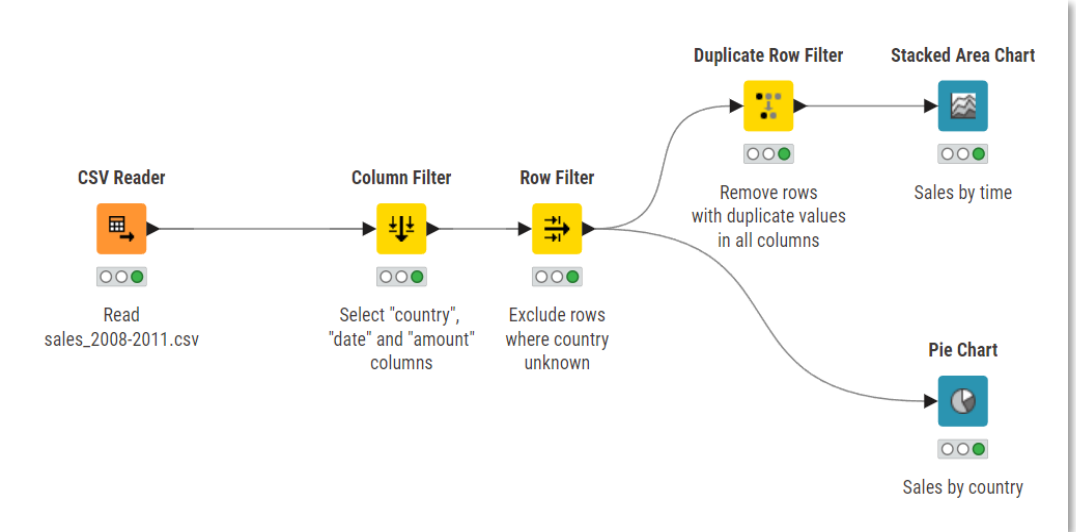
Nodes and Workflows

- **NODES** realizam tarefas com os dados



- Nodes são combinados para criar **WORKFLOWS**



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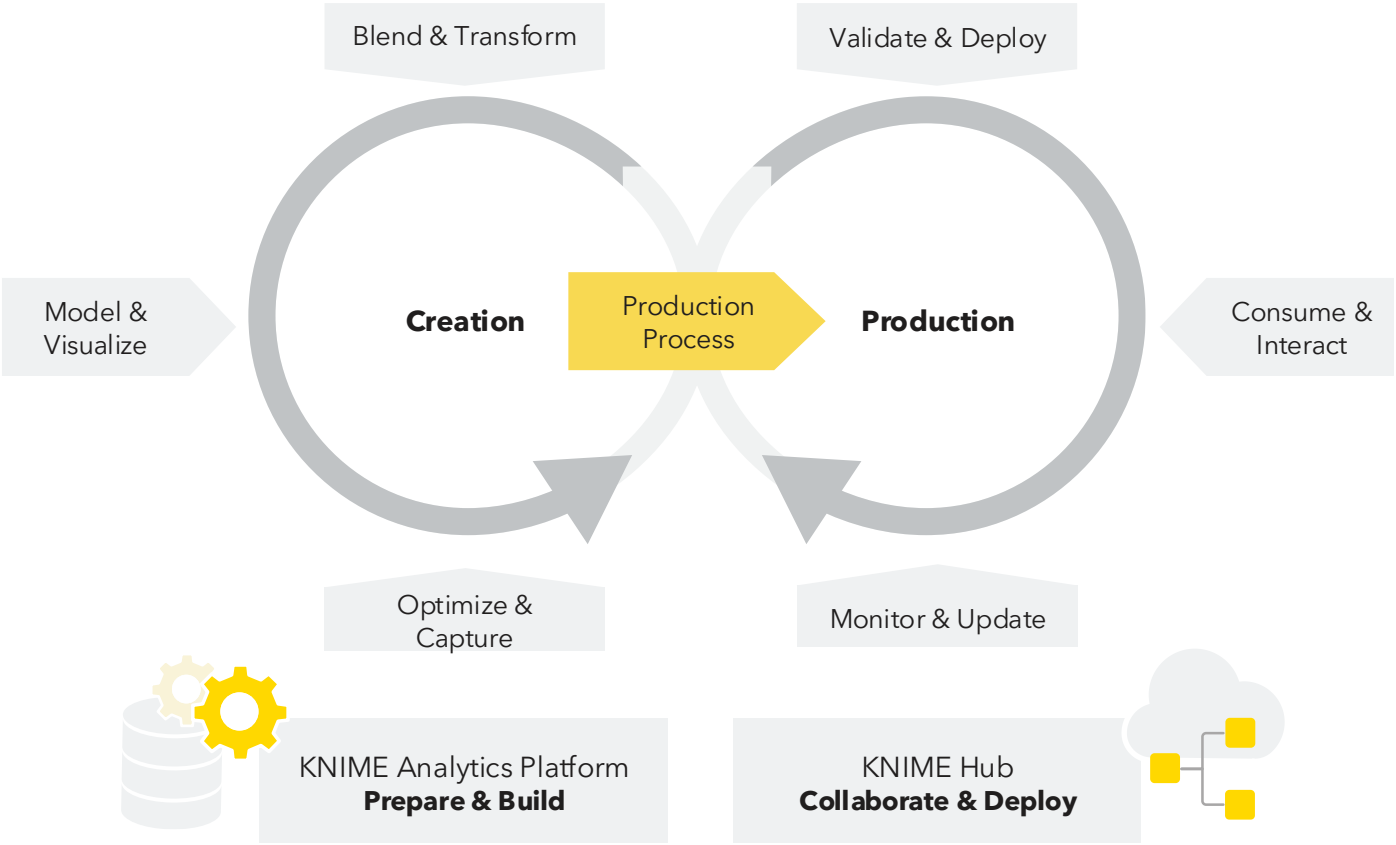
Ciclo de vida de Ciência de dados com o Knime



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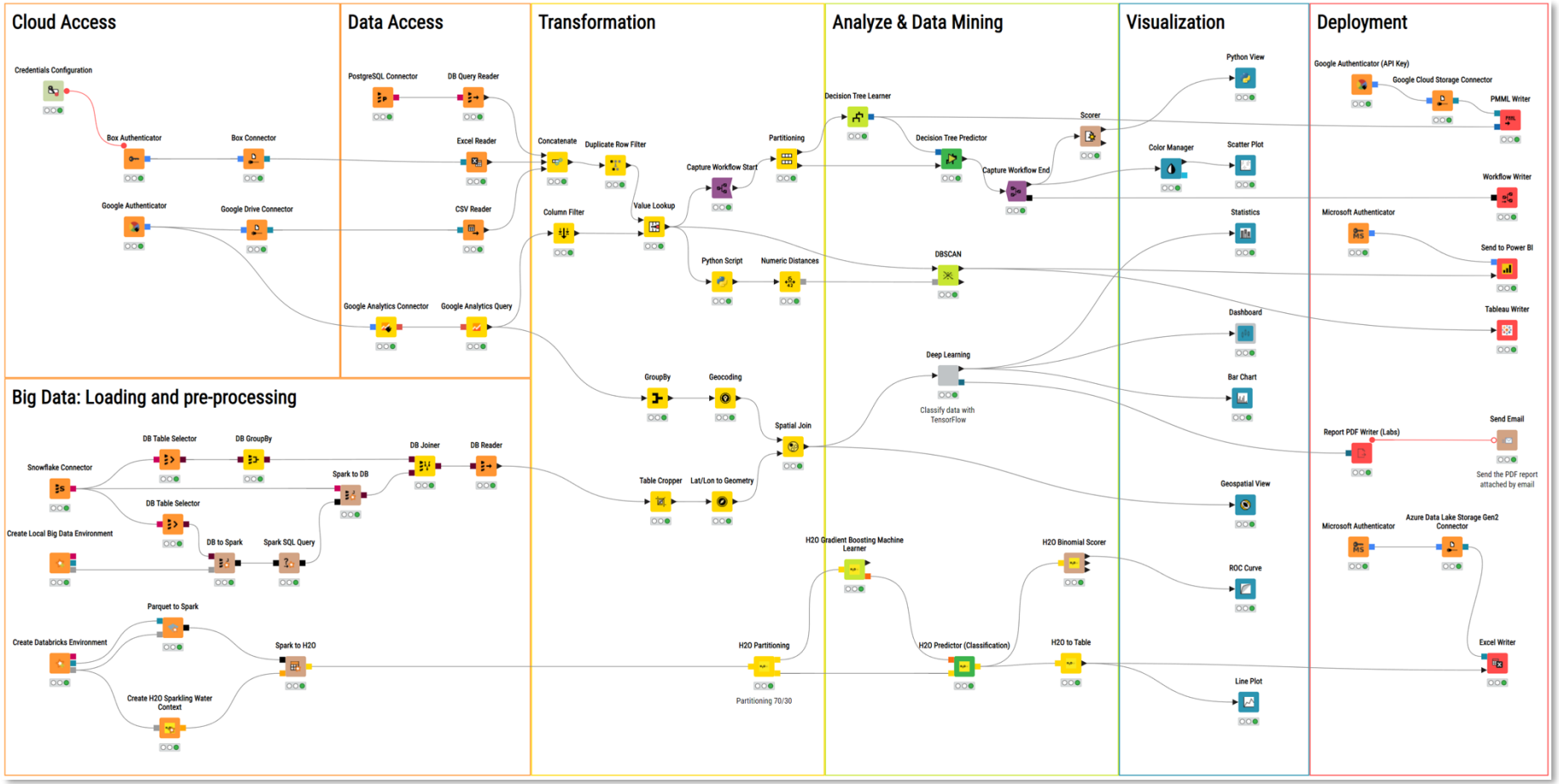
Mais de 4.000 Nodes incluso



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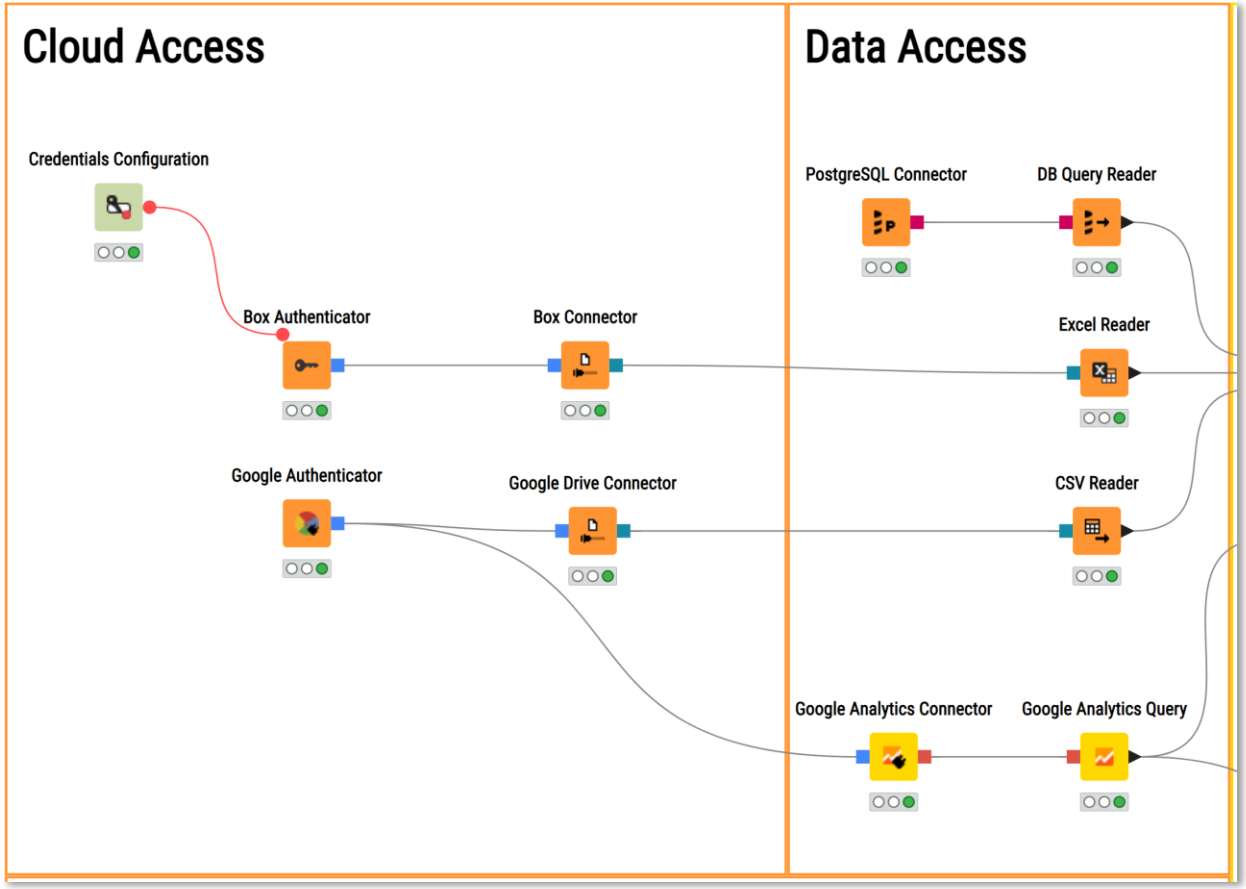
Acesso aos Dados: diferentes plataformas



- Files
  - CSV, txt, Excel, Word, PDF
  - SAS, SPSS
  - XML, JSON, PMML
  - Images, texts, networks
- Databases
  - MySQL, PostgreSQL, Oracle
  - Theobald
  - any JDBC (DB2, MS SQL Server)
  - Amazon DynamoDB
- Other
  - Twitter, Google
  - Amazon S3, Azure Blob Store
  - Sharepoint, Salesforce
  - Kafka
  - REST, Web services

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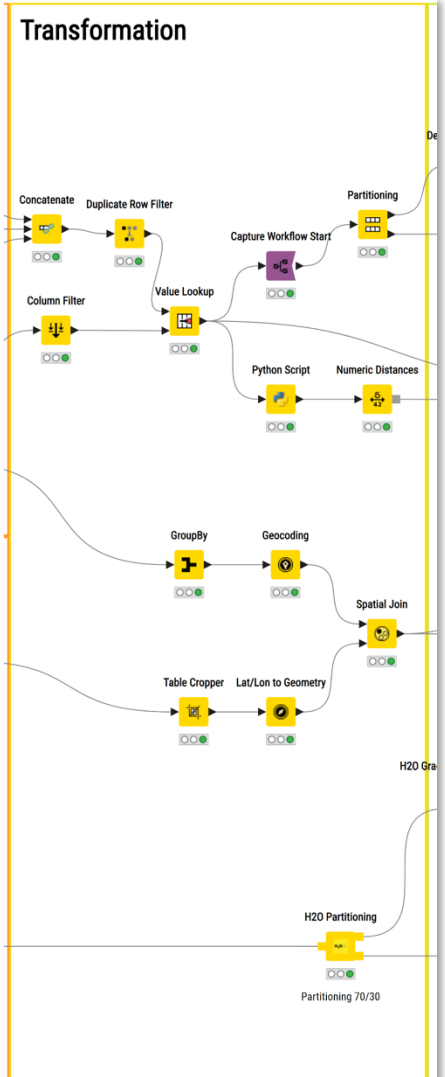
Transformação



- Limpeza de dados
  - Row, column, table based
- Junção de dados
  - Join, concatenate, append
- Agregação
  - Grouping, pivoting, binning
- Criação e seleção de características

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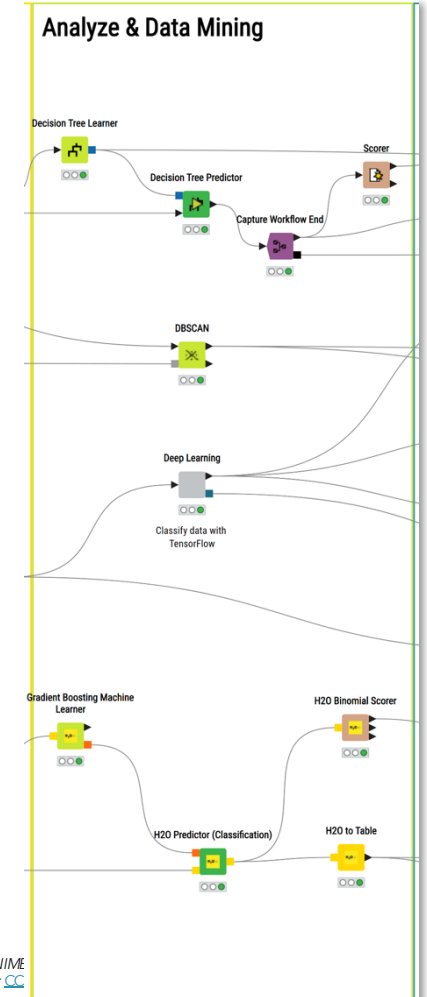
Análise & Mineração de dados



- Regressão
  - Linear, regression tree
- Classificação
  - Decision tree, ensembles, SVM, MLP, Naïve Bayes, logistic regression
- Clusterização
  - k-means, DBSCAN, hierarchical
- Validação
  - Cross-validation, scoring, ROC
- Deep Learning
  - Keras, DL4J
- Externo
  - R, Python, Weka, H2O, Keras

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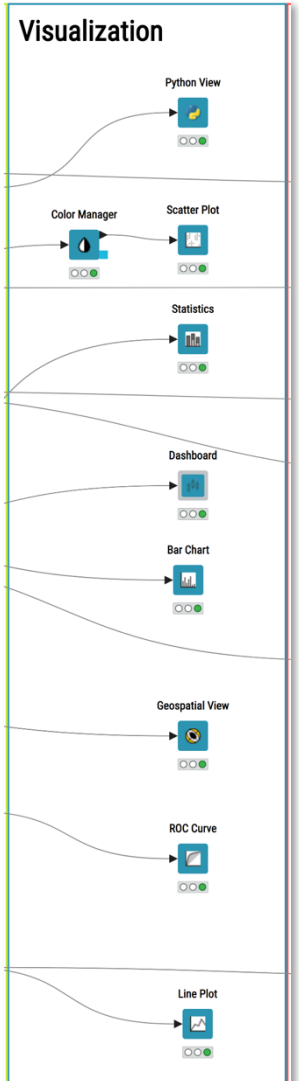
Visualização



- Interactive dashboards
- Many visualization nodes
  - Scatter Plot, Box Plot, Line Plot
  - Networks, ROC Curve, Decision Tree
  - Adding more with each release!
- Misc
  - Tag cloud, geospatial, molecules
- Script-based visualizations
  - R, Python, JavaScript

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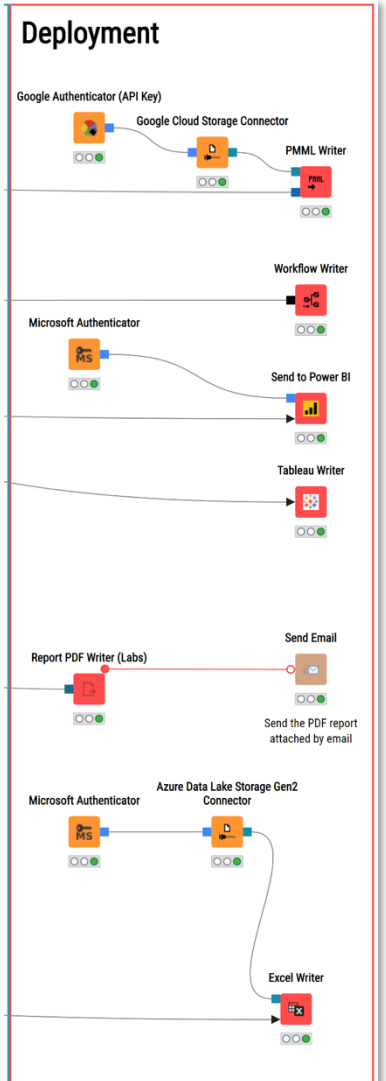
Entrega



- Database
- Files
  - Excel, CSV, txt
  - XML
  - PMML
  - to: local, KNIME Business Hub, Cloud
- Reporting

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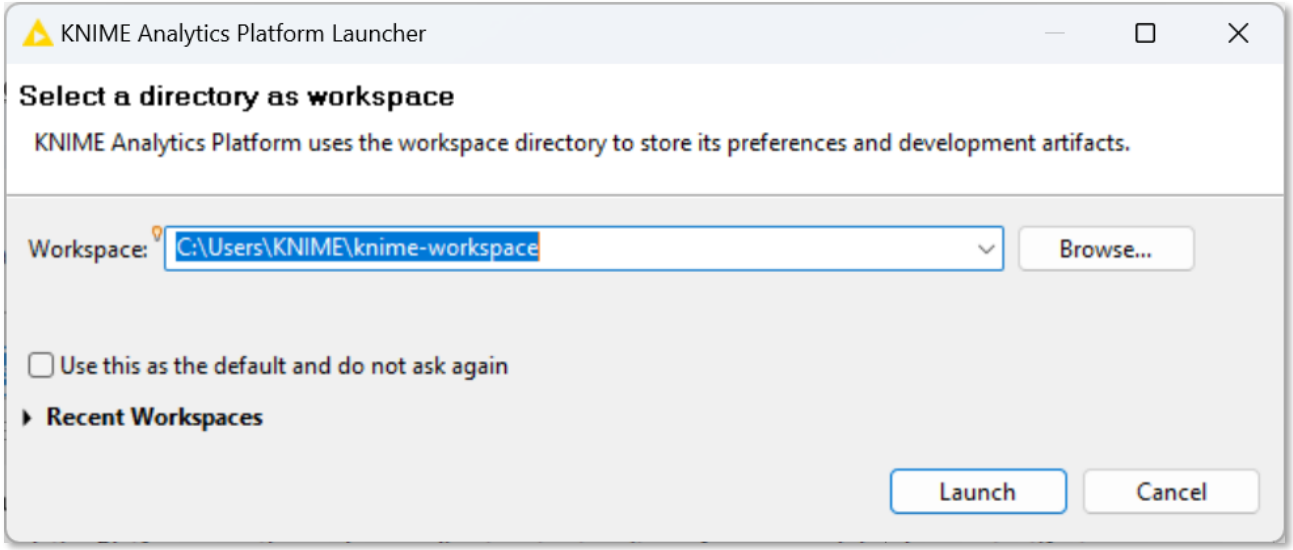
KNIME Workspace

- Pasta local onde os fluxos de trabalho e arquivos de dados são armazenados na sessão Corrente do Knime
  - Workspaces são portáveis, bem como o KNIME



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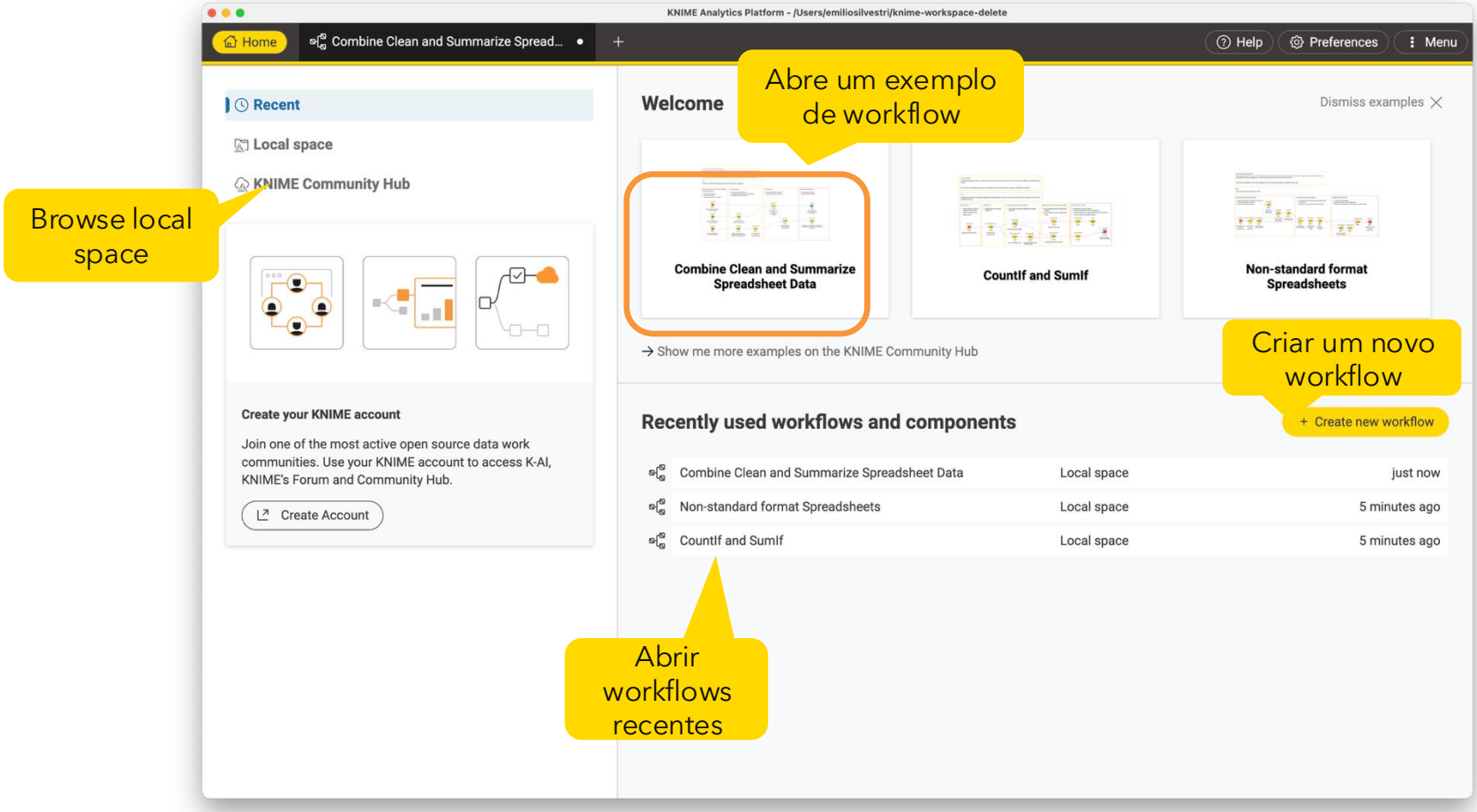
Tour da interface



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Tour da interface

Description tab

Application tabs

Workflow editor

Node monitor

Home | Combine Clean and Summarize Spread... | Help | Preferences | Menu

Execute | Cancel | Reset | Create metanode | Create component | 55%

Excel Reader

This node reads Excel files (xlsx, xlsm, xlsb, and xls format). It can read a single or multiple files at the same time, however reading only one sheet per file. The supported Excel types that can be read in are string, number, boolean, date, and time but not pictures, diagrams, etc.
The data is read in and converted to the KNIME types string, integer, long, double, boolean, local date, local time, and local date&time. Upon execution, the node will scan the input file to determine number and types of the columns and output a table with the auto-guessed structure and KNIME types.
Formulas can also be read in and reevaluated if desired.The performance of this node is limited (due to the underlying library of the Apache POI project). Reading large files takes a long time and uses a lot of memory (especially files in xlsx format when formula reevaluation is enabled).

The dialog of this node shows a *preview* and the *file content*. While the preview shows the table that will be read in having the settings applied and is updated once a setting is changed, the file content shows the content of the file as it is displayed in Excel. This allows finding row numbers and column names easier to specify different settings.

This node can access a variety of different file systems. More information about file handling in KNIME can be found in the official File Handling Guide.

Ports | Options | Views

1: File Table | Flow Variables

Rows: 9 | Columns: 2 | Table | Statistics

#	RowID	Item String	Amount Number (integer)
1	Row0	Table	1
2	Row1	Chair	2
3	Row2	Cupboard	1
4	Row3	Side table	1
5	Row4	Fridge	1

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Tour da interface

Repositório de Nodes

Buscar nodes

Drag and drop para adicioná-lo no workflow

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Tour da interface

Space Explorer tab

exibe workflows no seu local space

O space explorer abre arquivos na pasta do corrente workflow

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Tour da interface

Workflow Monitor tab

Click to highlight the failing node in the workflow

List error and warnings in the workflow

KNIME Analytics Platform - /Users/emiliosilvestri/knime-workspace-delete

Home | Combine Clean and Summarize Spread... | Help | Preferences | Menu

Execute | Cancel | 55%

Workflow monitor

Node errors

Excel Reader

Execute failed: The specified file Rooms1.xlsx does not exist.

1: File Table | Flow Variables

To show the port output, please execute the selected node.

Execute

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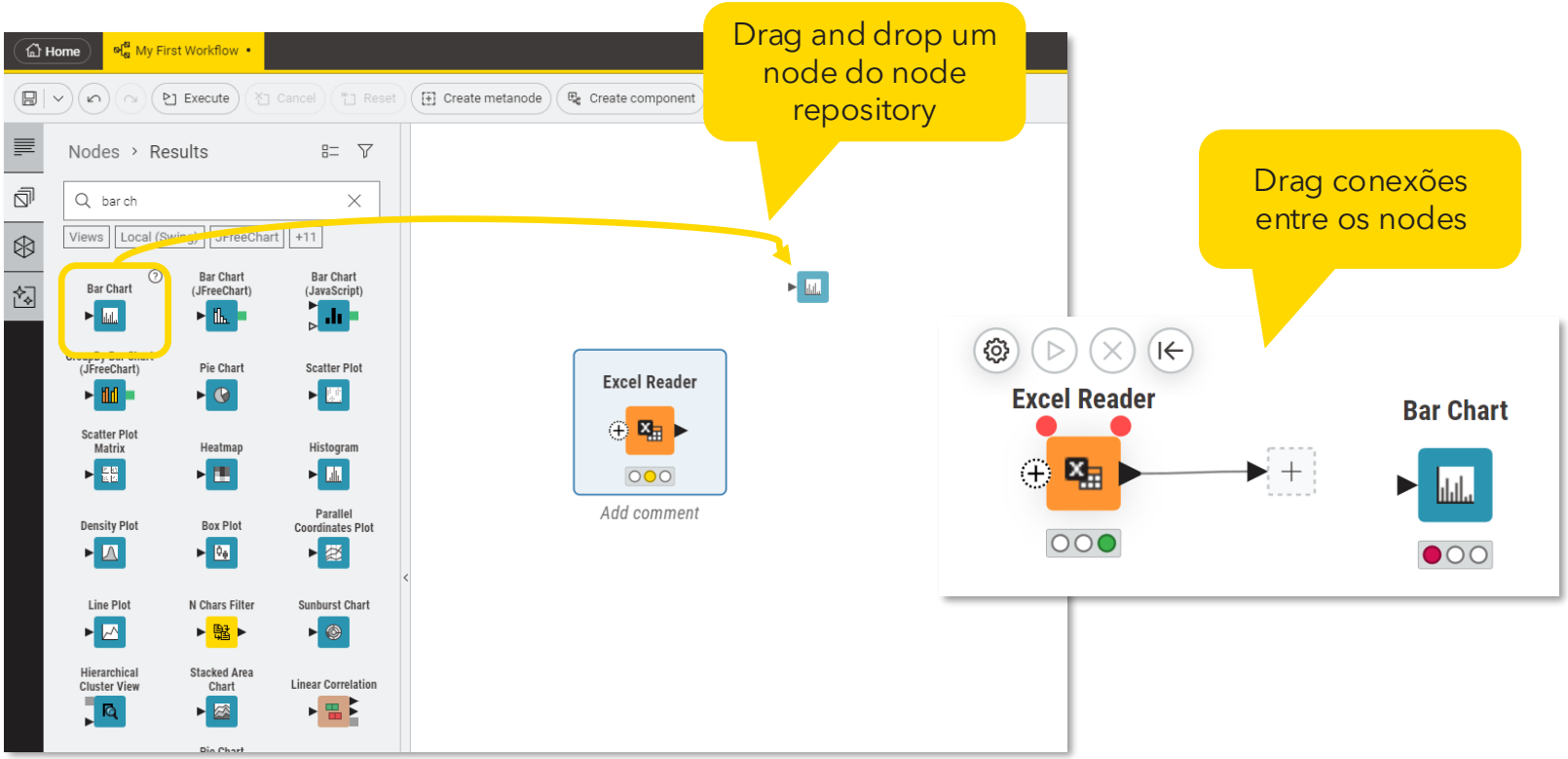
Adicionar e conectar Nodes



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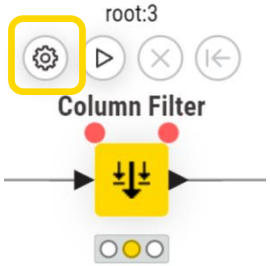
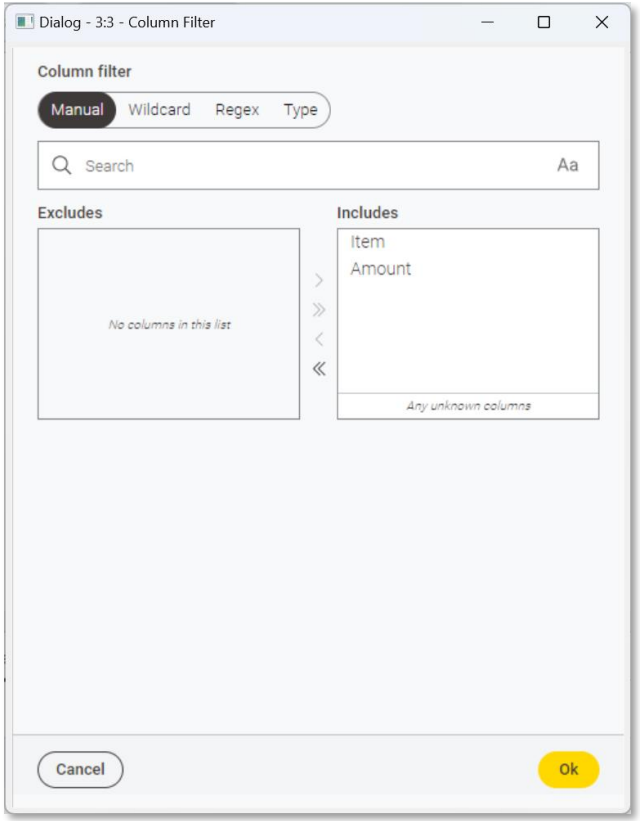
Configurar um Node

- muitos nodes requerem configuração
- Para abrir a Caixa de diálogo de configuração:
  - Duplo clique no node
  - Botão da direita-> Configure
  - Selecione o node e pressione F6
- Clique no botão Configurar na barra de ações do node





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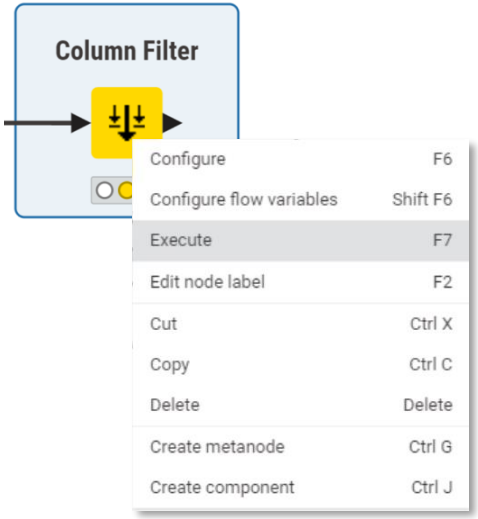
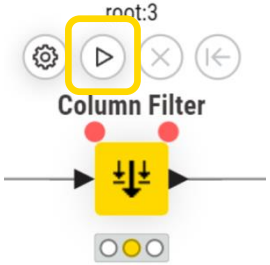
Executar um Node

- Um node executado realizou a sua tarefa
- Opções para executar um node:
  - Botão da direita-> Execute
  - Selecione o nó e pressione F7
  - Clique no botão Execute no barra de ações do nó





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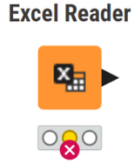
Node Status

- Um node pode ter 4 status:

Não configurado:
O node está aguardando a configuração ou a carga de dados

Configurado:
O node foi configurado e já pode ser executado

Executado:
O node foi executado com sucesso
Resultados pode ser visualizados e usados por outros nós



Erro:
Ocorreu um erro na execução do node

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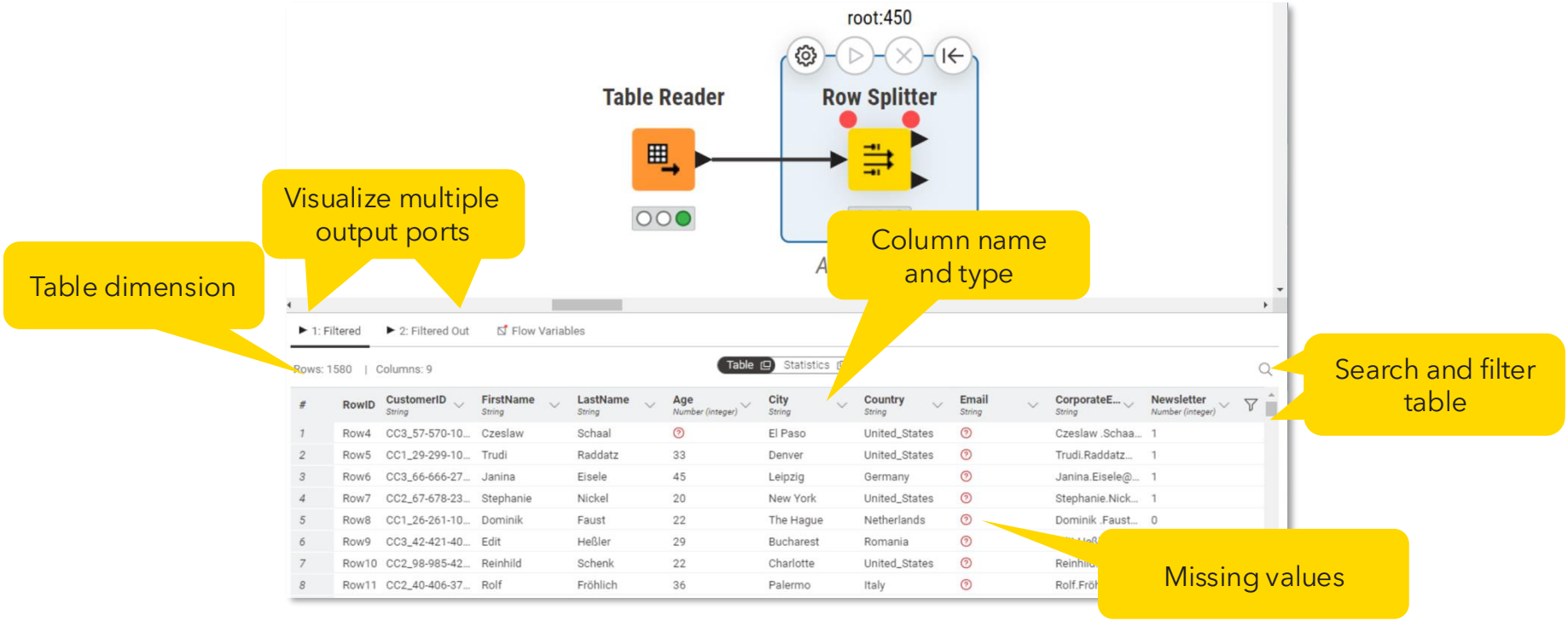
Node Monitor

- Selecione um nó já executado para inspecionar suas saídas



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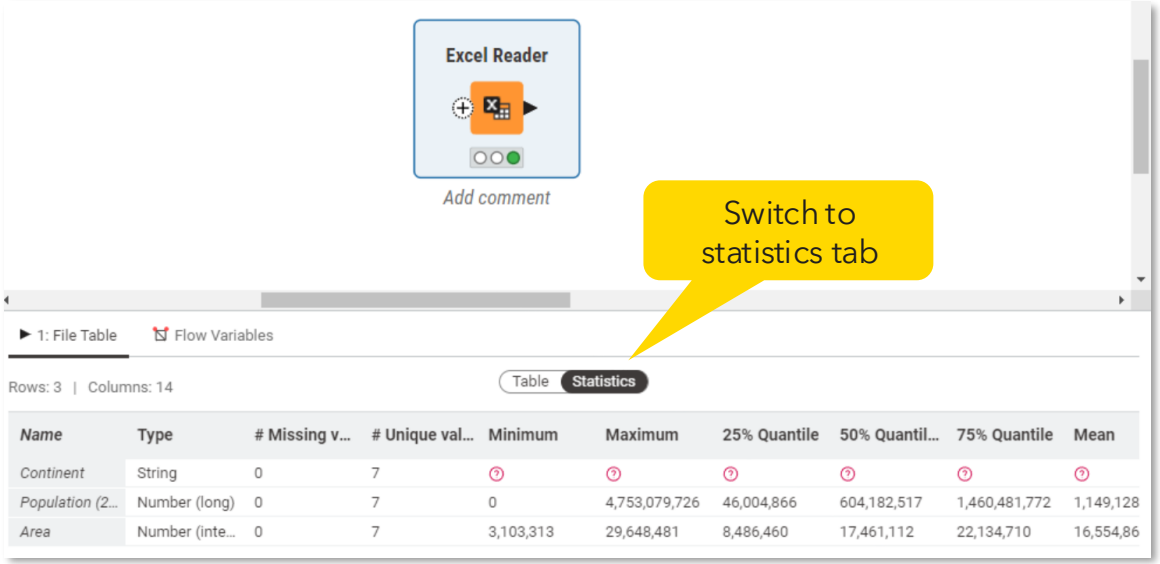
Statistics Tab

- Obtendo estatística da porta de saída do node
  - Missing values (dados ausentes)
  - Valores únicos
  - Valores mínimos e máximos
  - Quantis
  - …



Name	Type	# Missing v...	# Unique val...	Minimum	Maximum	25% Quantile	50% Quantil...	75% Quantile	Mean
Continent	String	0	7	?	?	?	?	?	?
Population (2...	Number (long)	0	7	0	4,753,079,726	46,004,866	604,182,517	1,460,481,772	1,149,128
Area	Number (inte...	0	7	3,103,313	29,648,481	8,486,460	17,461,112	22,134,710	16,554,86

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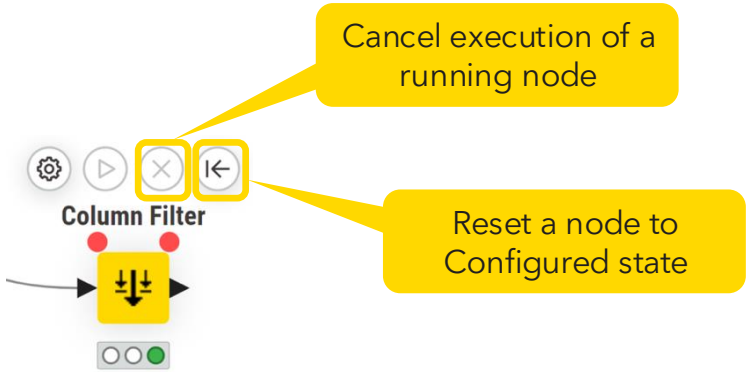
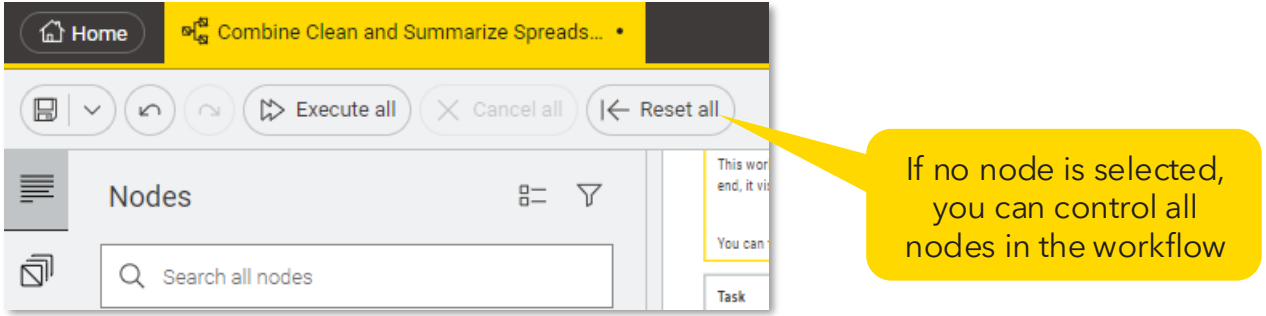
Cancelar e Resetar a Node



- O controle de Node também pode ser feitono Workflow Toolbar



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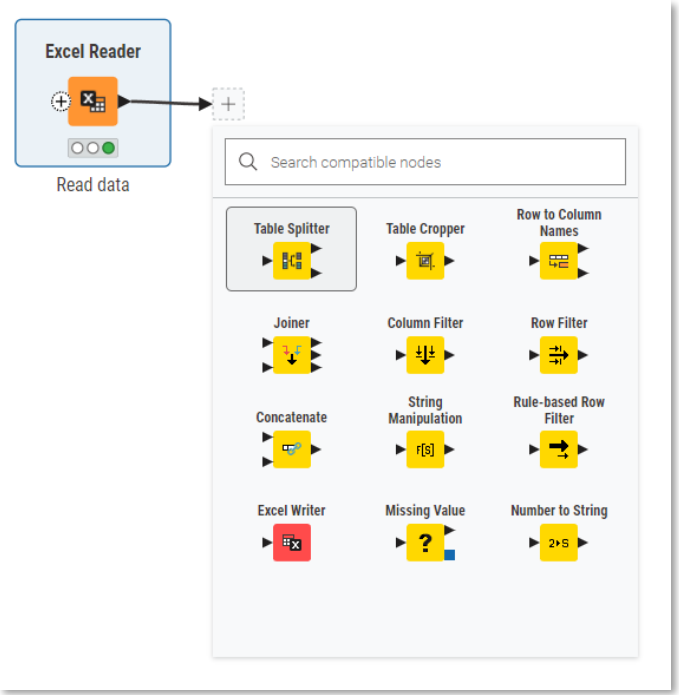
Quick Node Adding Panel

- Drag and drop na saída de um node para um espaço vazio
- Veja a recomendação de nodes
- Procure por todos nodes compatíveis



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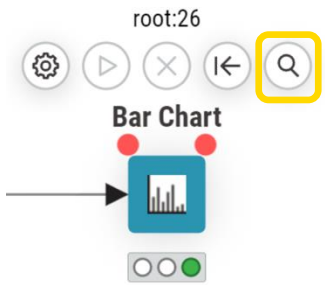
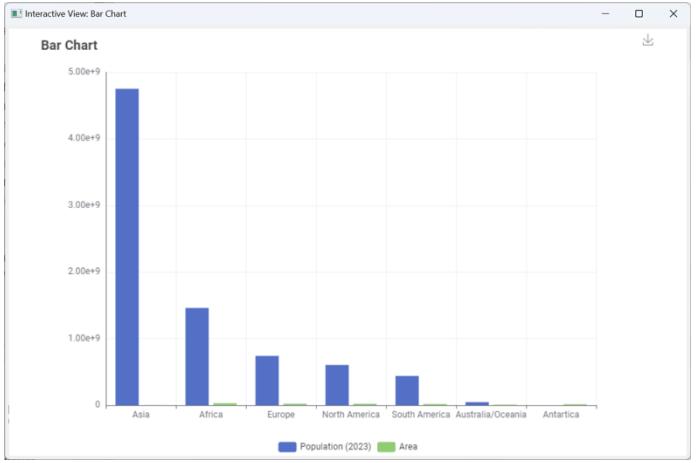
Node Views

- Abre o node view em uma nova janela
- Somente disponível para “blue” nodes





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Extensões de arquivos KNIME

Workflow_1.knwf

- ***.knwf** for KNIME Workflow Files

Uso: Representa **um único workflow** criado no KNIME.
Quando usar:
Ao exportar apenas um workflow específico para backup, compartilhamento ou migração.
Para enviar um projeto individual para outro usuário abrir e executar.
Conteúdo: Contém todos os nós, configurações e metadados daquele workflow.

- ***.knar** for KNIME Archive Files (folder)

Group_wf_1.knar

•**Uso**: Representa **um arquivo compactado** que pode conter:
  •Múltiplos workflows
  •Workflows com suas dependências
  •Metadados adicionais (por exemplo, estrutura de projeto)
•**Quando usar**:
  •Para exportar um **conjunto de workflows** ou todo um **workspace**.
  •Para transferir ou fazer backup de vários projetos de uma só vez.
•**Conteúdo**: Estrutura hierárquica completa com workflows, subpastas e metainformações.

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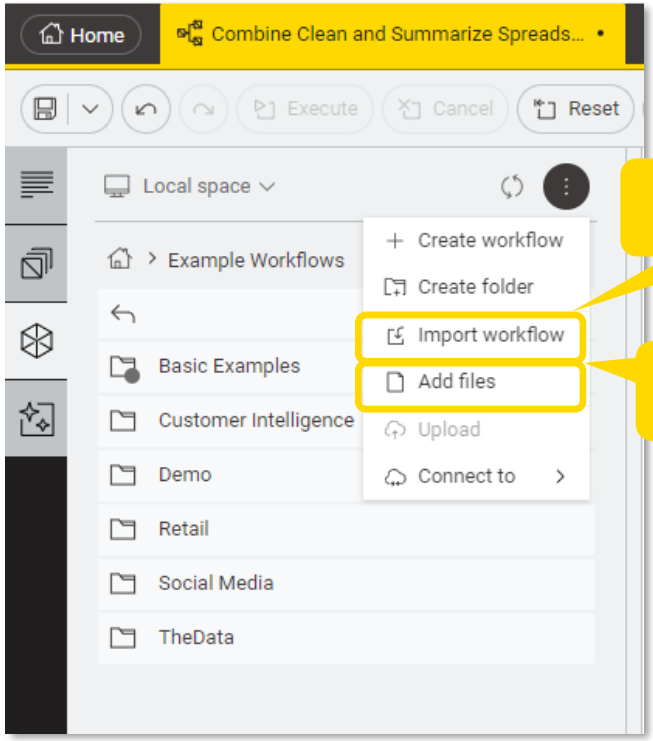
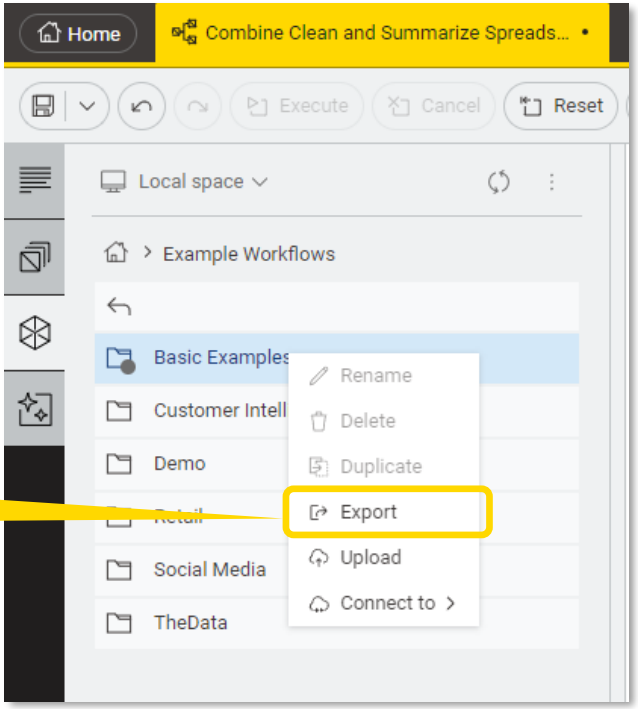
Importação e Exportação de Workflows



Import .knwf or .knar files

Import data files

Export to file



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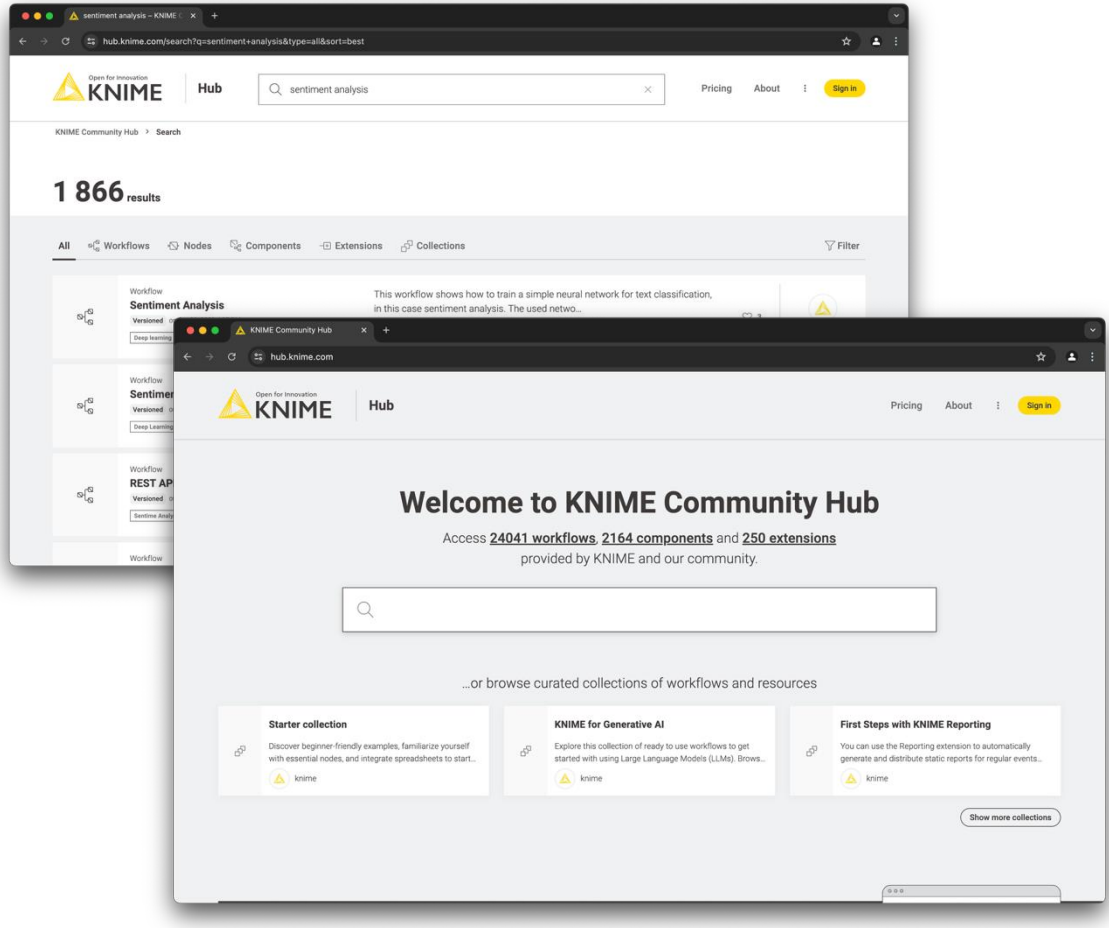
KNIME Community Hub

- Espaço para encontrar e compartilhar
  - Workflows
  - Nodes
  - Components
  - Extensions

- Espaço de colaboração para equipes
- Executa e distribui workflows



hub.knime.com

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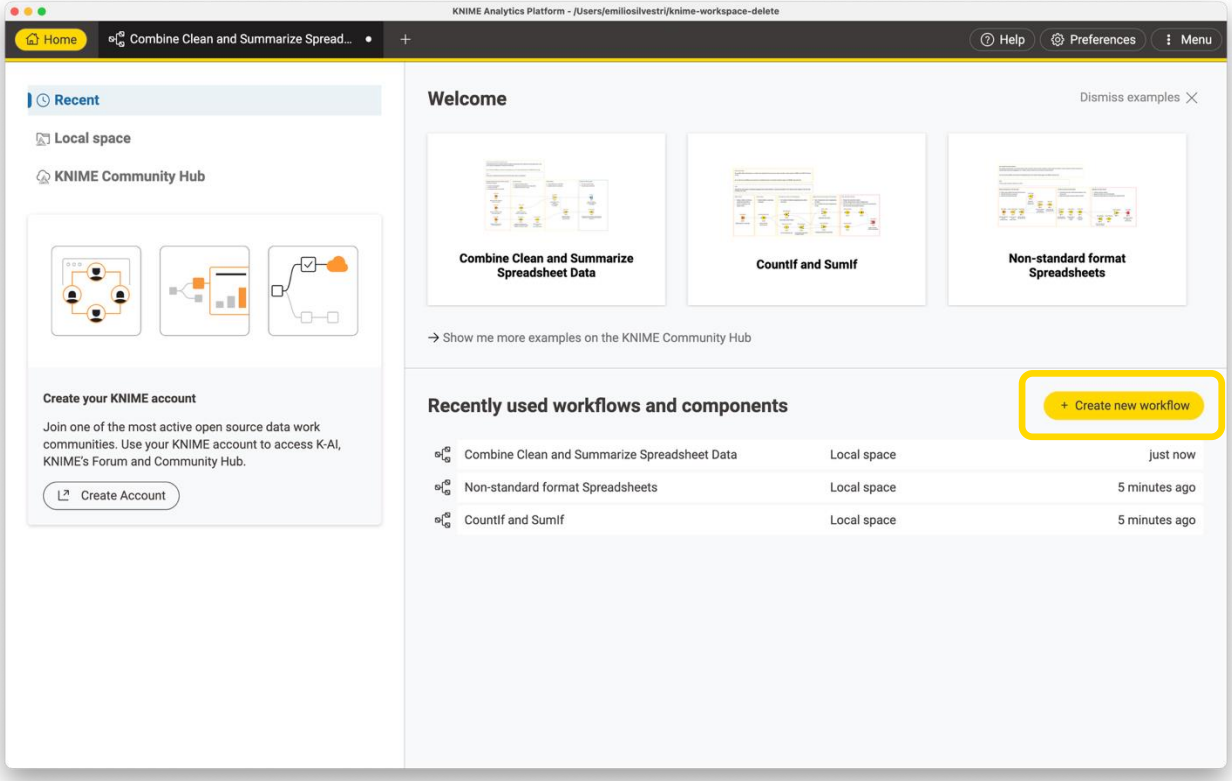
O primeiro Knime Workflow a gente nunca esquece…

Construindo seu primeiro Workflow

- Na página Home, crie um novo workflow e forneça um nome



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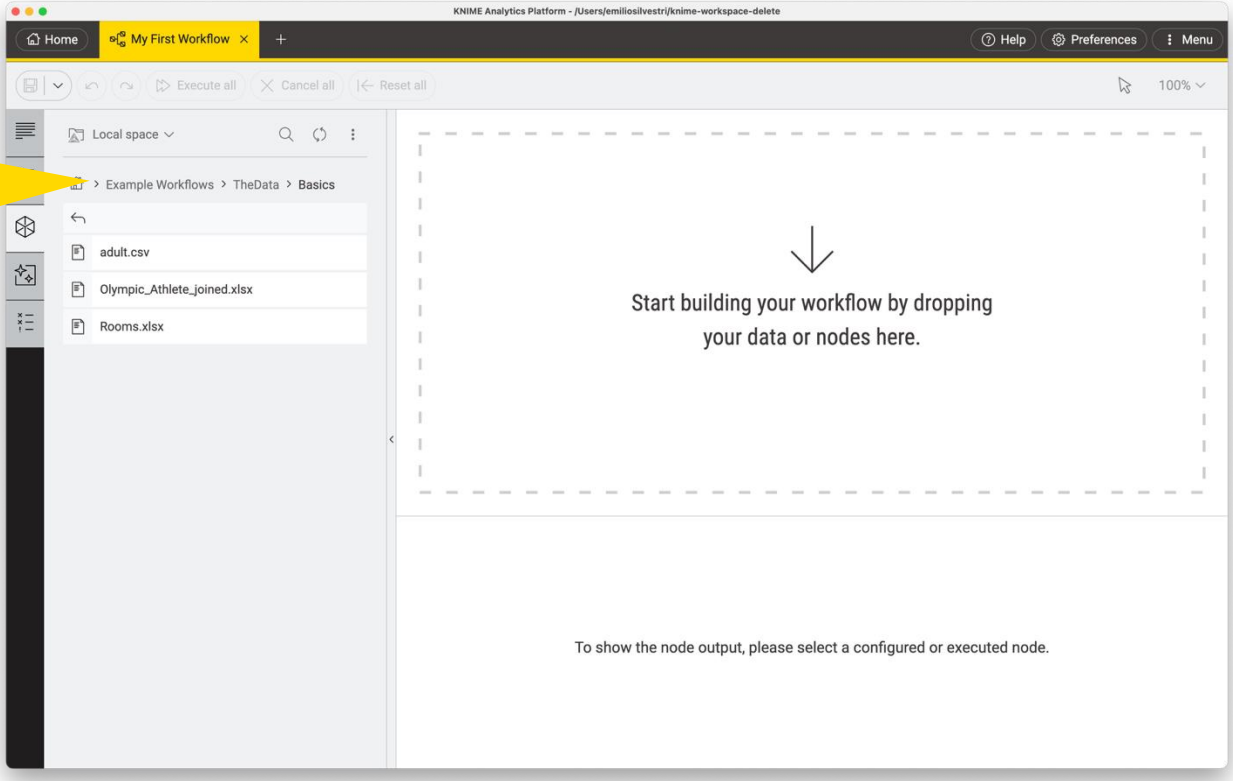
Construindo seu primeiro Workflow

- No painel de navegação, abra a Guia Space Explorer

Navigate to Example Workflows > TheData > Basics



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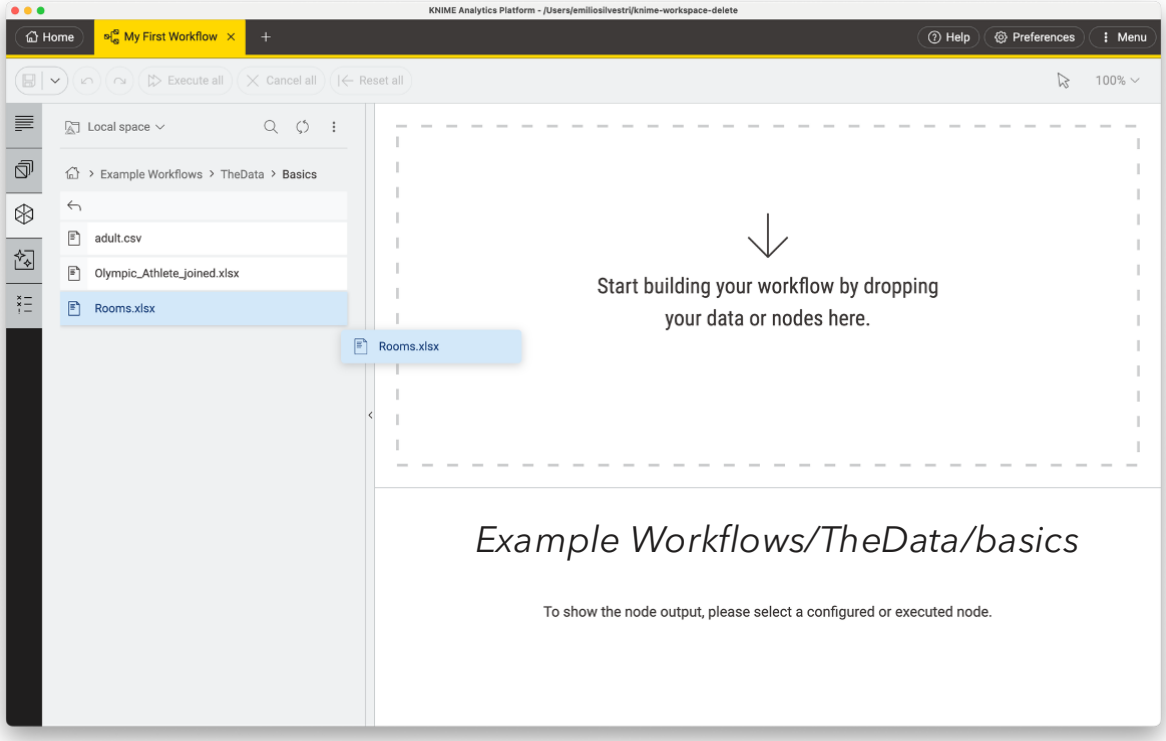
Construindo seu primeiro Workflow

- Arraste e solte o arquivo **Rooms.xlsx** dentro da área vazia do Workflow Editor



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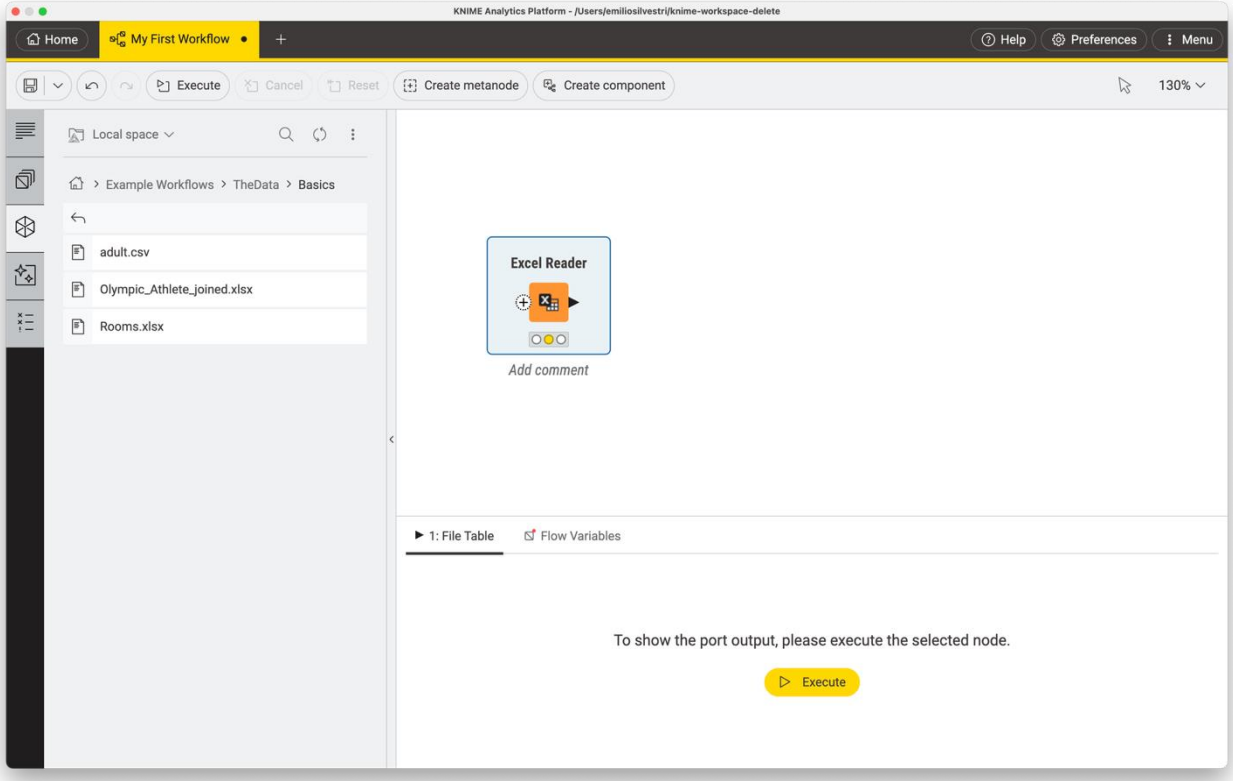
Construindo seu primeiro Workflow

- Um Excel Reader node surgirá no Workflow Editor



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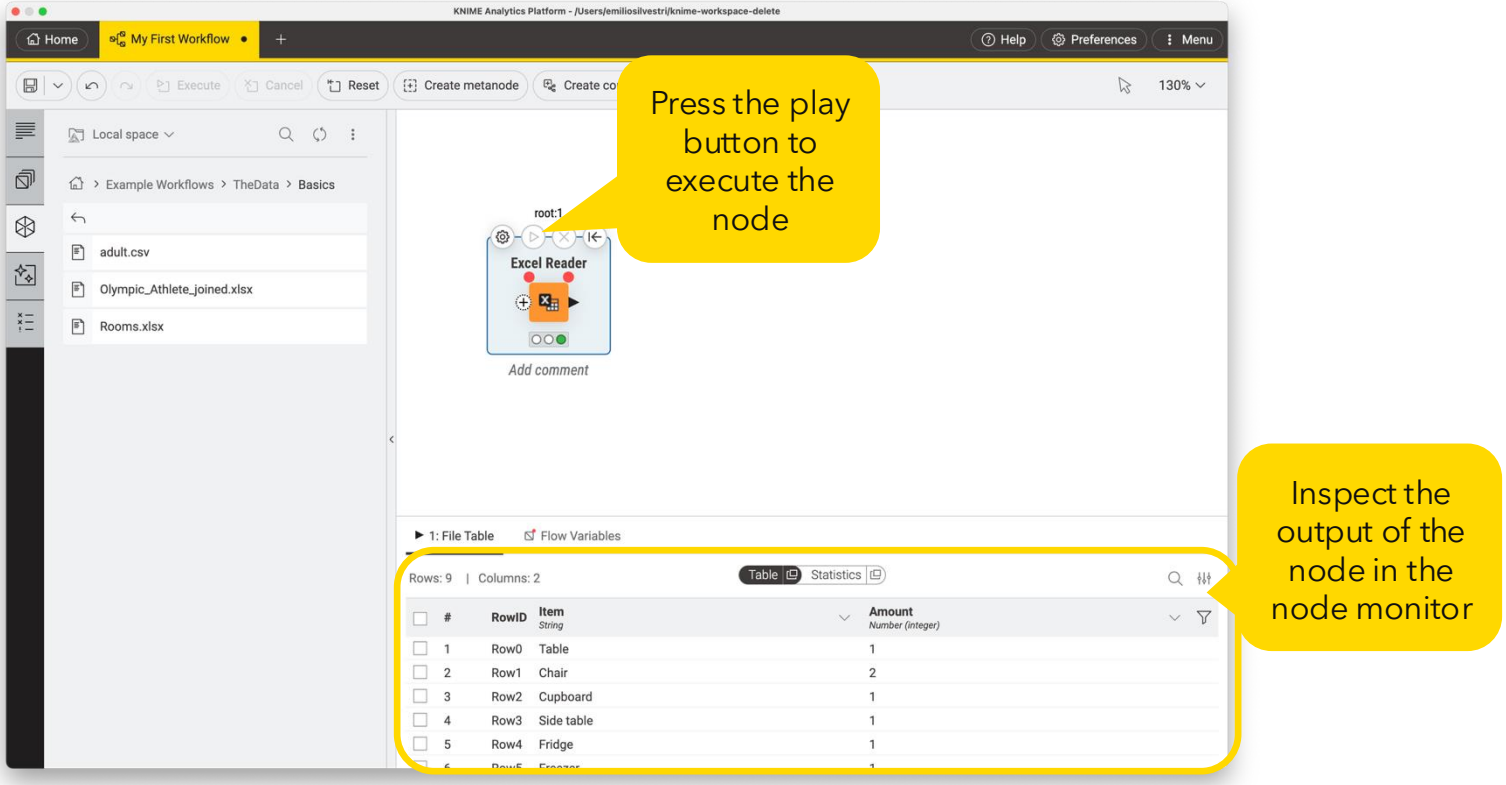
Construindo seu primeiro Workflow

- Selecione o node e execute



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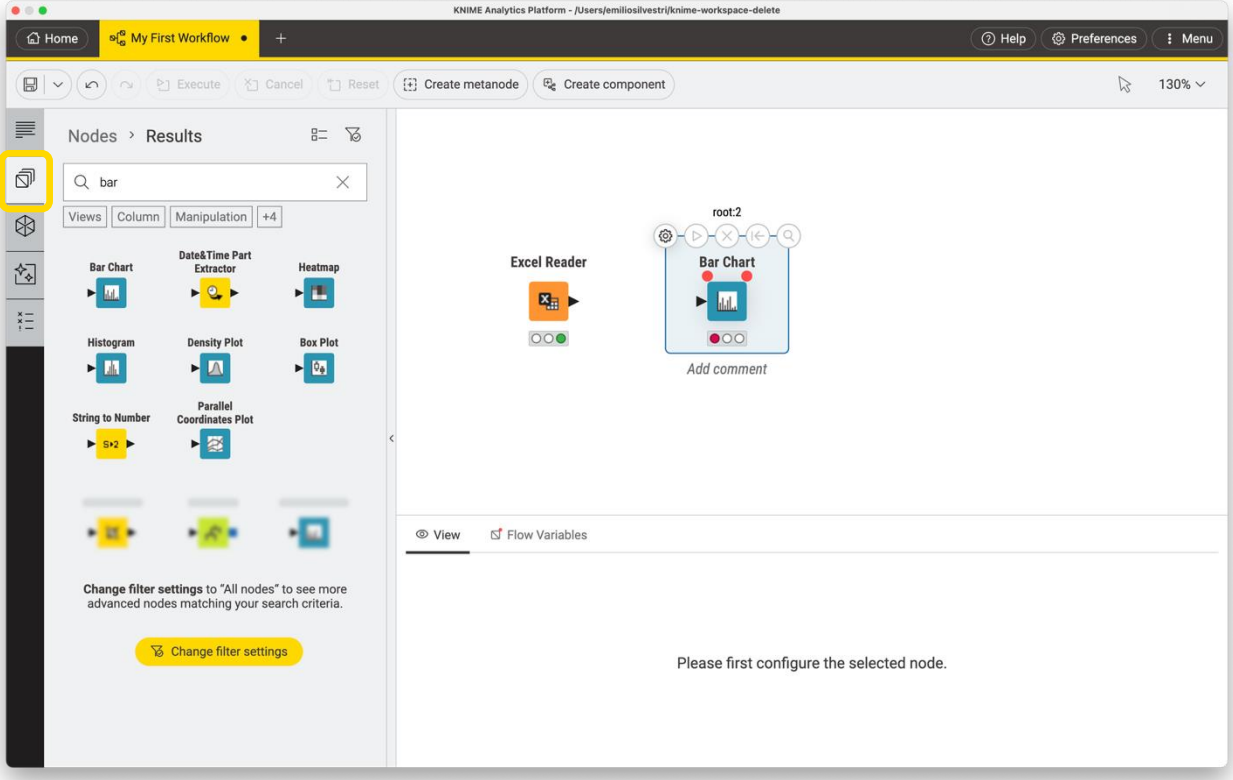
Construindo seu primeiro Workflow

- Na guia de repostório de Nodes, arraste e solte Bar Chart node



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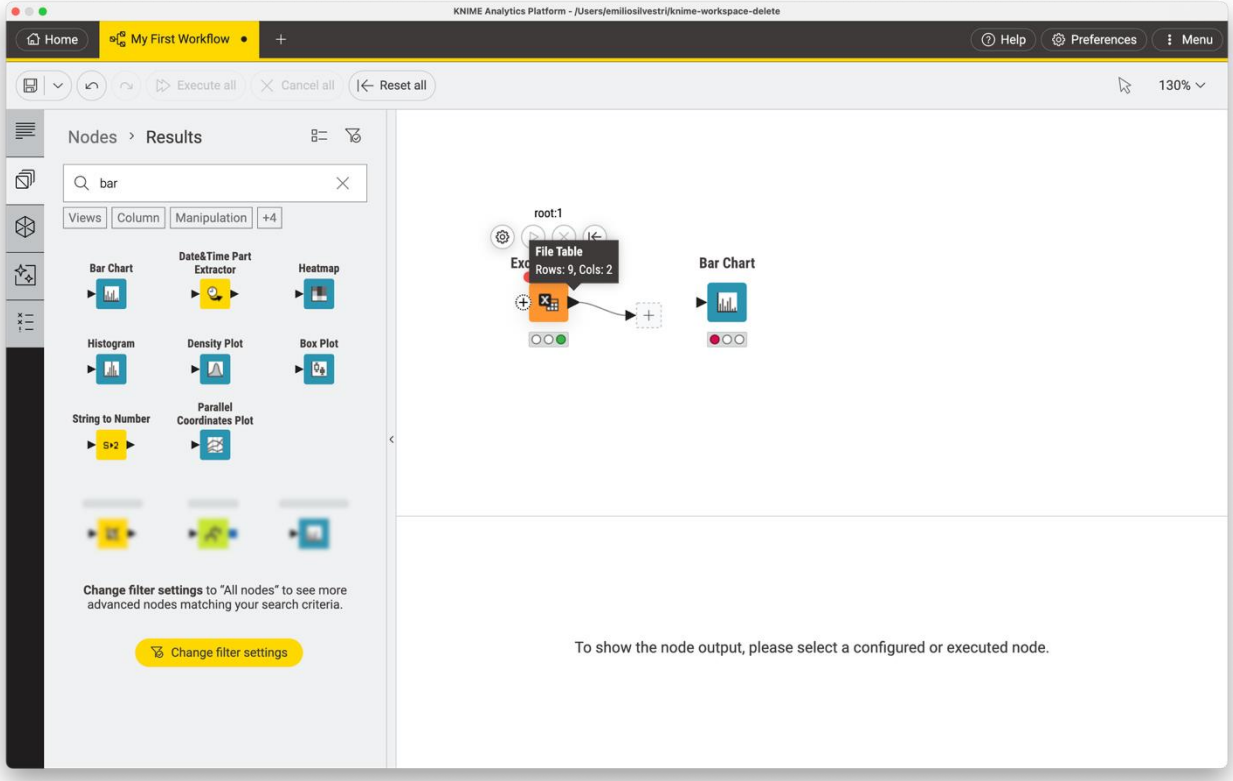
Construindo seu primeiro Workflow

- Arraste o conector entre os dois nodes



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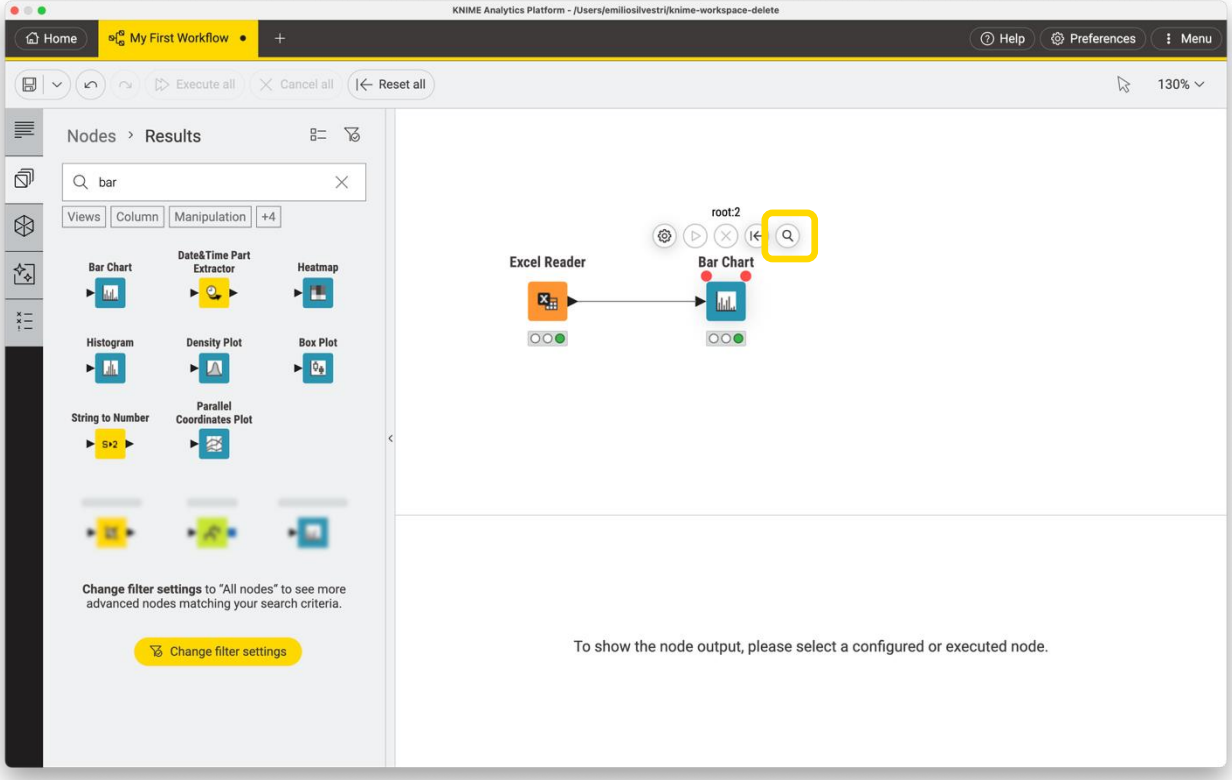
Construindo seu primeiro Workflow

- Passe o cursor sobre o gráfico de barras e clique no ícone de lupa



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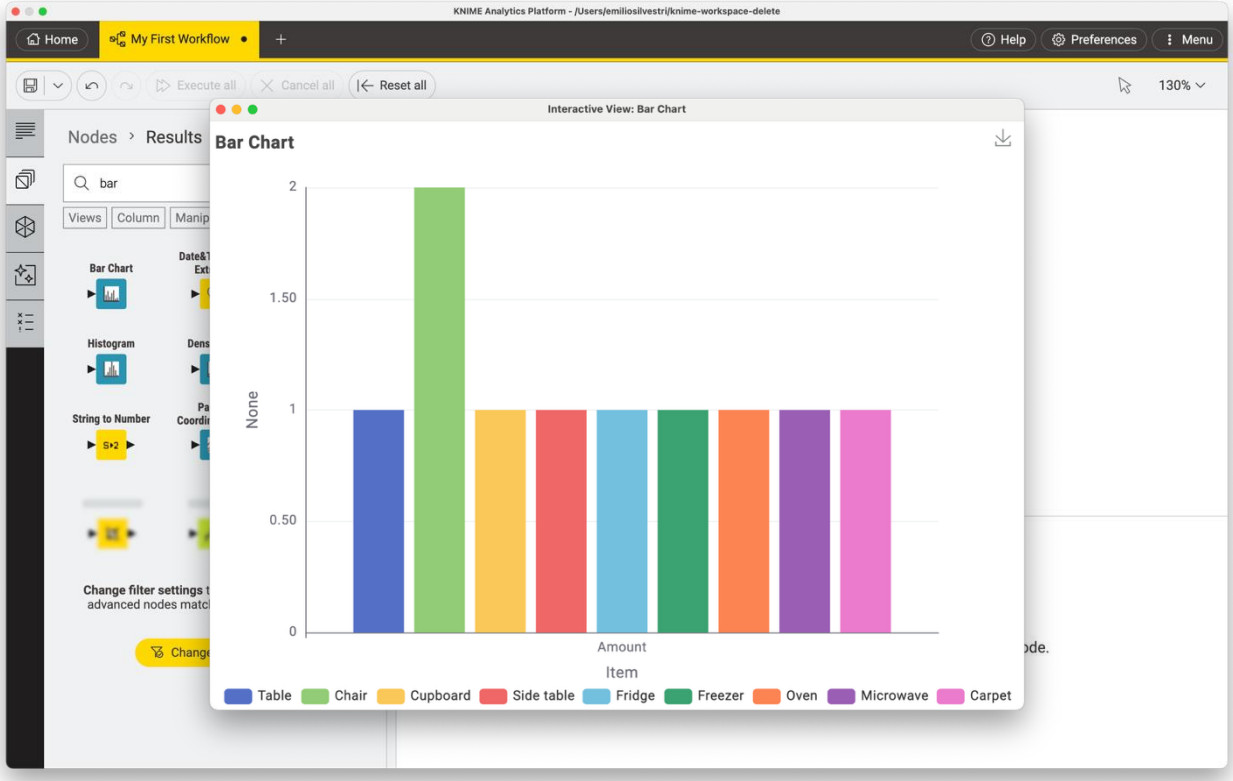
Construindo seu primeiro Workflow

- Este é a visualização de seu primeiro fluxo de trabalho:



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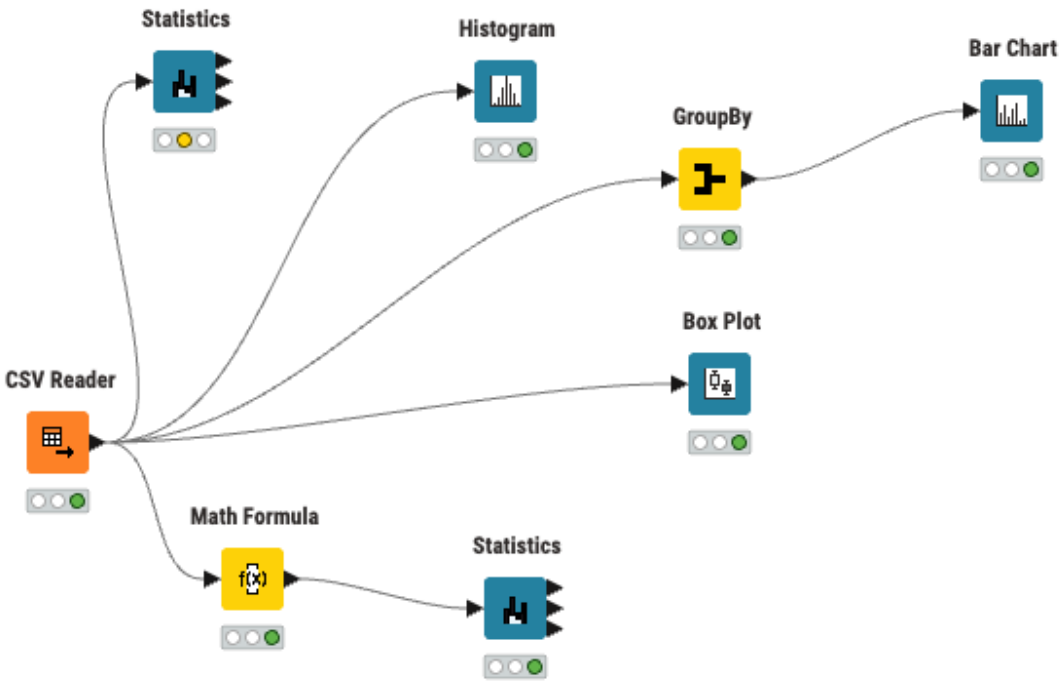
Atividade 2: Análise Estatística do Perfil Biométrico de Pacientes

Objetivo: Fazer a análise exploratória dos dados de 1.000 pacientes para entender a distribuição de idade, peso e altura, além de identificar padrões por tipo sanguíneo.



Análise estatística com Knime

- **Passo 1: Carga de Dados**
- Arraste o nó **CSV Reader** para o workflow.
- Configure-o para ler o arquivo **dados_pacientes.csv.**

- **Passo 2: Estatística Geral (Tendência Central e Dispersão)**
- Conecte um nó **Statistics** à saída do CSV Reader.
- Execute o nó e abra a visualização (**View: Statistics View**).

- **Tarefa:** Identifique a média, a mediana (50% quantile) e o desvio padrão da coluna peso(kg).
  - *Pergunta:* O peso médio é muito diferente da mediana?
  - Isso indica alguma assimetria nos dados?

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Análise estatística com Knime

- **Passo 3: Visualização de Distribuição**
- Conecte um nó **Histogram** ao CSV Reader.
  - Nas configurações, selecione a coluna idade(anos).
  - **Tarefa:** Observe se a idade dos pacientes está uniformemente distribuída ou se há uma concentração em alguma faixa etária específica.

- **Passo 4: Análise de Agrupamento (Média por Categoria)**
- Conecte um nó **GroupBy** ao CSV Reader.
- Na aba **Groups**, selecione a coluna tipo_sanguineo.
- Na aba **Manual Aggregation**, adicione a coluna peso(kg) e escolha o método **Mean** (Média).
  - **Tarefa:** Verifique qual tipo sanguíneo possui a maior média de peso.

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Análise estatística com Knime

- **Passo 5: Identificação de Outliers**
- Conecte um nó **Box Plot** ao CSV Reader.
- Selecione a coluna altura(cm).
- **Tarefa:** Verifique no gráfico se existem "pontos" fora dos limites (whiskers), que indicariam pacientes com alturas muito acima ou abaixo da média esperada.

- **Passo 6 : Cálculo do IMC**
- Adicione um nó **Math Formula**.
- Crie uma nova coluna chamada IMC.
- Use a fórmula: PESO / ((ALTURA/100) * (ALTURA/100)).
- Conecte um novo nó de **Statistics** para ver a média de IMC desta população.

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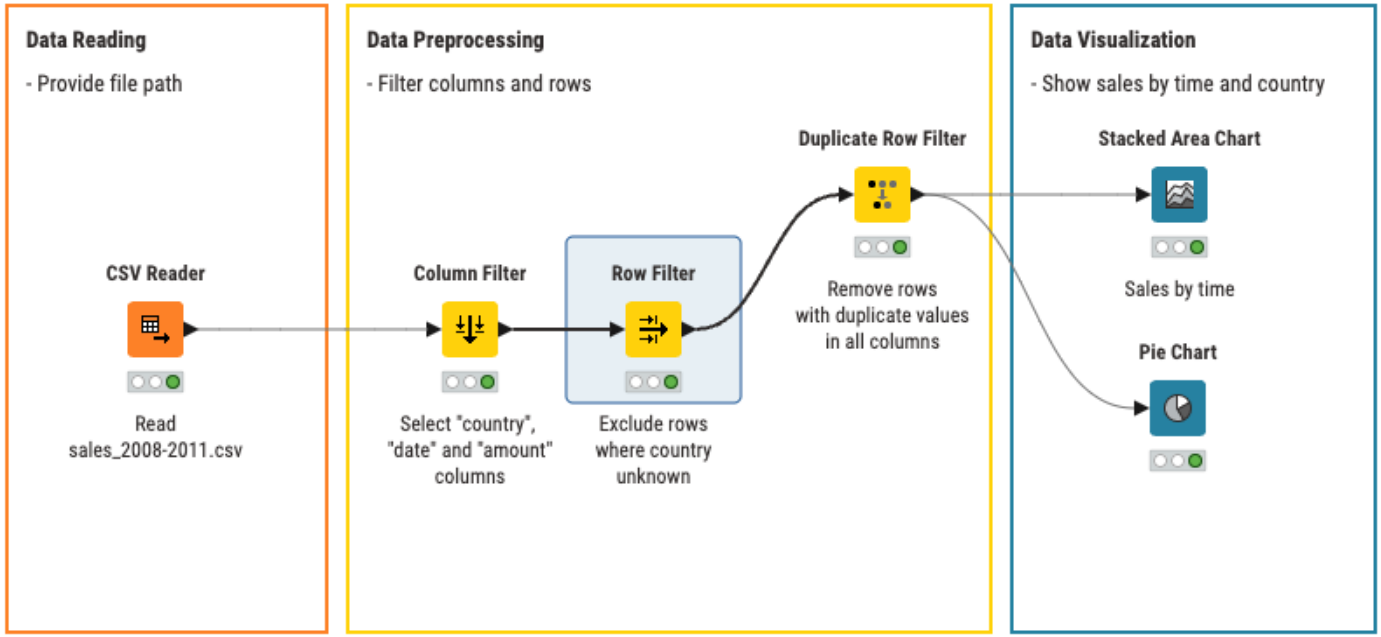
Atividade 3

Use: sales_2008-2011.csv